

GTOS Detailed Volume - Volume 17

Testing Verification Conformance

GTOS Enterprise Engineering Handbook - Final Edition 1.0

Source Chapters

- Testing and Conformance Introduction
- Unit Testing
- Property Testing
- Aggregate Invariant Testing
- Contract Testing
- Testcontainers Integration
- Database Migration Verification
- Architecture Tests
- Tenant Isolation Tests
- Security Adversarial Tests
- Event Replay Tests
- Workflow Restart and Compensation Tests
- AI Evaluation
- Portal E2E Journeys
- Accessibility Testing
- Performance and Capacity Testing
- Resilience and Fault Injection
- Disaster Recovery Exercises
- Partner Certification
- Continuous Conformance
- Volume 17 Conformance and Evidence Matrix

Testing and Conformance Introduction

Metadata	Value
Volume	17
Chapter ID	V17-00
Subject	TestingandConformanceIntroduction
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Testing and Conformance Introduction.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
TestingandConformanceIntroduction	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TestingandConformanceIntroductionVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TestingandConformanceIntroductionDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TestingandConformanceIntroductionEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TestingandConformanceIntroductionException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
TestingandConformanceIntroduction {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
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```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateTestingandConformanceIntroduction
- ValidateTestingandConformanceIntroduction
- ApproveTestingandConformanceIntroduction
- ActivateTestingandConformanceIntroduction
- SuspendTestingandConformanceIntroduction
- CorrectTestingandConformanceIntroduction
- RetireTestingandConformanceIntroduction

Reference Events:

- TestingandConformanceIntroductionCreated
- TestingandConformanceIntroductionValidated
- TestingandConformanceIntroductionApproved
- TestingandConformanceIntroductionActivated
- TestingandConformanceIntroductionSuspended
- TestingandConformanceIntroductionCorrected
- TestingandConformanceIntroductionRetired

Reference queries:

- GetTestingandConformanceIntroduction
- ListTestingandConformanceIntroductionRecords
- GetTestingandConformanceIntroductionTimeline
- GetTestingandConformanceIntroductionEvidence
- GetTestingandConformanceIntroductionDecisionPackage
- GetTestingandConformanceIntroductionExceptions
- GetTestingandConformanceIntroductionAuditTrail

Reference API surface:

```

POST /api/v1/testingandconformanceintroduction/commands
GET  /api/v1/testingandconformanceintroduction/{id}
GET  /api/v1/testingandconformanceintroduction/{id}/timeline
GET  /api/v1/testingandconformanceintroduction/{id}/evidence
GET  /api/v1/testingandconformanceintroduction/{id}/decisions
POST /api/v1/testingandconformanceintroduction/{id}/exceptions
POST /api/v1/testingandconformanceintroduction/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
testingandconformanceintroduction_aggregate	authoritative subject
testingandconformanceintroduction_version	immutable submitted, approved, issued, or signed versions
testingandconformanceintroduction_relationship	governed relationships
testingandconformanceintroduction_decision	binding Decisions and rationale
testingandconformanceintroduction_evidence_link	provenance and evidence relationships
testingandconformanceintroduction_event_outbox	transactional Event publication
testingandconformanceintroduction_idempotency	duplicate-write protection
testingandconformanceintroduction_exception	exception, claim, dispute, or hold
testingandconformanceintroduction_correction	correction and supersession lineage
testingandconformanceintroduction_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions

- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Testing and Conformance Introduction is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Unit Testing

Metadata	Value
Volume	17
Chapter ID	V17-01
Subject	UnitTesting
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
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1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Unit Testing**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
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- ≠ Recommendation
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1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
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- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
UnitTesting	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
UnitTestingVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
UnitTestingDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
UnitTestingEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
UnitTestingException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
UnitTesting {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
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- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
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```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
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```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateUnitTesting
- ValidateUnitTesting
- ApproveUnitTesting
- ActivateUnitTesting
- SuspendUnitTesting
- CorrectUnitTesting
- RetireUnitTesting

Reference Events:

- UnitTestingCreated
- UnitTestingValidated
- UnitTestingApproved
- UnitTestingActivated
- UnitTestingSuspended
- UnitTestingCorrected
- UnitTestingRetired

Reference queries:

- GetUnitTesting
- ListUnitTestingRecords
- GetUnitTestingTimeline
- GetUnitTestingEvidence
- GetUnitTestingDecisionPackage
- GetUnitTestingExceptions
- GetUnitTestingAuditTrail

Reference API surface:

```

POST /api/v1/unittesting/commands
GET  /api/v1/unittesting/{id}
GET  /api/v1/unittesting/{id}/timeline
GET  /api/v1/unittesting/{id}/evidence
GET  /api/v1/unittesting/{id}/decisions
POST /api/v1/unittesting/{id}/exceptions
POST /api/v1/unittesting/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
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unittesting_evidence_link	provenance and evidence relationships
unittesting_event_outbox	transactional Event publication
unittesting_idempotency	duplicate-write protection
unittesting_exception	exception, claim, dispute, or hold
unittesting_correction	correction and supersession lineage
unittesting_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
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Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

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Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

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13. Testing and Continuous Conformance

- aggregate invariant and transition tests
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- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
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- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
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Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

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uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
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No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Property Testing

Metadata	Value
Volume	17
Chapter ID	V17-02
Subject	PropertyTesting
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Property Testing**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
PropertyTesting	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PropertyTestingVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PropertyTestingDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PropertyTestingEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PropertyTestingException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
PropertyTesting {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreatePropertyTesting
- ValidatePropertyTesting
- ApprovePropertyTesting
- ActivatePropertyTesting
- SuspendPropertyTesting
- CorrectPropertyTesting
- RetirePropertyTesting

Reference Events:

- PropertyTestingCreated
- PropertyTestingValidated
- PropertyTestingApproved
- PropertyTestingActivated
- PropertyTestingSuspended
- PropertyTestingCorrected
- PropertyTestingRetired

Reference queries:

- GetPropertyTesting
- ListPropertyTestingRecords
- GetPropertyTestingTimeline
- GetPropertyTestingEvidence
- GetPropertyTestingDecisionPackage
- GetPropertyTestingExceptions
- GetPropertyTestingAuditTrail

Reference API surface:

```

POST /api/v1/propertytesting/commands
GET  /api/v1/propertytesting/{id}
GET  /api/v1/propertytesting/{id}/timeline
GET  /api/v1/propertytesting/{id}/evidence
GET  /api/v1/propertytesting/{id}/decisions
POST /api/v1/propertytesting/{id}/exceptions
POST /api/v1/propertytesting/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
propertytesting_aggregate	authoritative subject
propertytesting_version	immutable submitted, approved, issued, or signed versions
propertytesting_relationship	governed relationships
propertytesting_decision	binding Decisions and rationale
propertytesting_evidence_link	provenance and evidence relationships
propertytesting_event_outbox	transactional Event publication
propertytesting_idempotency	duplicate-write protection
propertytesting_exception	exception, claim, dispute, or hold
propertytesting_correction	correction and supersession lineage
propertytesting_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Property Testing is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Aggregate Invariant Testing

Metadata	Value
Volume	17
Chapter ID	V17-03
Subject	AggregateInvariantTesting
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Aggregate Invariant Testing**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
AggregateInvariantTesting	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AggregateInvariantTestingVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AggregateInvariantTestingDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AggregateInvariantTestingEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AggregateInvariantTestingException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
AggregateInvariantTesting {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateAggregateInvariantTesting
- ValidateAggregateInvariantTesting
- ApproveAggregateInvariantTesting
- ActivateAggregateInvariantTesting
- SuspendAggregateInvariantTesting
- CorrectAggregateInvariantTesting
- RetireAggregateInvariantTesting

Reference Events:

- AggregateInvariantTestingCreated
- AggregateInvariantTestingValidated
- AggregateInvariantTestingApproved
- AggregateInvariantTestingActivated
- AggregateInvariantTestingSuspended
- AggregateInvariantTestingCorrected
- AggregateInvariantTestingRetired

Reference queries:

- GetAggregateInvariantTesting
- ListAggregateInvariantTestingRecords
- GetAggregateInvariantTestingTimeline
- GetAggregateInvariantTestingEvidence
- GetAggregateInvariantTestingDecisionPackage
- GetAggregateInvariantTestingExceptions
- GetAggregateInvariantTestingAuditTrail

Reference API surface:

```

POST /api/v1/aggregateinvarianttesting/commands
GET  /api/v1/aggregateinvarianttesting/{id}
GET  /api/v1/aggregateinvarianttesting/{id}/timeline
GET  /api/v1/aggregateinvarianttesting/{id}/evidence
GET  /api/v1/aggregateinvarianttesting/{id}/decisions
POST /api/v1/aggregateinvarianttesting/{id}/exceptions
POST /api/v1/aggregateinvarianttesting/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
aggregateinvarianttesting_aggregate	authoritative subject
aggregateinvarianttesting_version	immutable submitted, approved, issued, or signed versions
aggregateinvarianttesting_relationship	governed relationships
aggregateinvarianttesting_decision	binding Decisions and rationale
aggregateinvarianttesting_evidence_link	provenance and evidence relationships
aggregateinvarianttesting_event_outbox	transactional Event publication
aggregateinvarianttesting_idempotency	duplicate-write protection
aggregateinvarianttesting_exception	exception, claim, dispute, or hold
aggregateinvarianttesting_correction	correction and supersession lineage
aggregateinvarianttesting_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Aggregate Invariant Testing is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Contract Testing

Metadata	Value
Volume	17
Chapter ID	V17-04
Subject	ContractTesting
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Contract Testing**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ContractTesting	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ContractTestingVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ContractTestingDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ContractTestingEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ContractTestingException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ContractTesting {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```



```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateContractTesting
- ValidateContractTesting
- ApproveContractTesting
- ActivateContractTesting
- SuspendContractTesting
- CorrectContractTesting
- RetireContractTesting

Reference Events:

- ContractTestingCreated
- ContractTestingValidated
- ContractTestingApproved
- ContractTestingActivated
- ContractTestingSuspended
- ContractTestingCorrected
- ContractTestingRetired

Reference queries:

- GetContractTesting
- ListContractTestingRecords
- GetContractTestingTimeline
- GetContractTestingEvidence
- GetContractTestingDecisionPackage
- GetContractTestingExceptions
- GetContractTestingAuditTrail

Reference API surface:

```

POST /api/v1/contracttesting/commands
GET  /api/v1/contracttesting/{id}
GET  /api/v1/contracttesting/{id}/timeline
GET  /api/v1/contracttesting/{id}/evidence
GET  /api/v1/contracttesting/{id}/decisions
POST /api/v1/contracttesting/{id}/exceptions
POST /api/v1/contracttesting/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
contracttesting_aggregate	authoritative subject
contracttesting_version	immutable submitted, approved, issued, or signed versions
contracttesting_relationship	governed relationships
contracttesting_decision	binding Decisions and rationale
contracttesting_evidence_link	provenance and evidence relationships
contracttesting_event_outbox	transactional Event publication
contracttesting_idempotency	duplicate-write protection
contracttesting_exception	exception, claim, dispute, or hold
contracttesting_correction	correction and supersession lineage
contracttesting_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Contract Testing is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Testcontainers Integration

Metadata	Value
Volume	17
Chapter ID	V17-05
Subject	TestcontainersIntegration
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Testcontainers Integration**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
TestcontainersIntegration	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TestcontainersIntegrationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TestcontainersIntegrationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TestcontainersIntegrationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TestcontainersIntegrationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
TestcontainersIntegration {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateTestcontainersIntegration
- ValidateTestcontainersIntegration
- ApproveTestcontainersIntegration
- ActivateTestcontainersIntegration
- SuspendTestcontainersIntegration
- CorrectTestcontainersIntegration
- RetireTestcontainersIntegration

Reference Events:

- TestcontainersIntegrationCreated
- TestcontainersIntegrationValidated
- TestcontainersIntegrationApproved
- TestcontainersIntegrationActivated
- TestcontainersIntegrationSuspended
- TestcontainersIntegrationCorrected
- TestcontainersIntegrationRetired

Reference queries:

- GetTestcontainersIntegration
- ListTestcontainersIntegrationRecords
- GetTestcontainersIntegrationTimeline
- GetTestcontainersIntegrationEvidence
- GetTestcontainersIntegrationDecisionPackage
- GetTestcontainersIntegrationExceptions
- GetTestcontainersIntegrationAuditTrail

Reference API surface:

```

POST /api/v1/testcontainersintegration/commands
GET  /api/v1/testcontainersintegration/{id}
GET  /api/v1/testcontainersintegration/{id}/timeline
GET  /api/v1/testcontainersintegration/{id}/evidence
GET  /api/v1/testcontainersintegration/{id}/decisions
POST /api/v1/testcontainersintegration/{id}/exceptions
POST /api/v1/testcontainersintegration/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```



```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
testcontainersintegration_aggregate	authoritative subject
testcontainersintegration_version	immutable submitted, approved, issued, or signed versions
testcontainersintegration_relationship	governed relationships
testcontainersintegration_decision	binding Decisions and rationale
testcontainersintegration_evidence_link	provenance and evidence relationships
testcontainersintegration_event_outbox	transactional Event publication
testcontainersintegration_idempotency	duplicate-write protection
testcontainersintegration_exception	exception, claim, dispute, or hold
testcontainersintegration_correction	correction and supersession lineage
testcontainersintegration_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Testcontainers Integration is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Database Migration Verification

Metadata	Value
Volume	17
Chapter ID	V17-06
Subject	DatabaseMigrationVerification
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Database Migration Verification**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DatabaseMigrationVerification	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseMigrationVerificationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseMigrationVerificationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseMigrationVerificationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseMigrationVerificationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DatabaseMigrationVerification {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDatabaseMigrationVerification
- ValidateDatabaseMigrationVerification
- ApproveDatabaseMigrationVerification
- ActivateDatabaseMigrationVerification
- SuspendDatabaseMigrationVerification
- CorrectDatabaseMigrationVerification
- RetireDatabaseMigrationVerification

Reference Events:

- DatabaseMigrationVerificationCreated
- DatabaseMigrationVerificationValidated
- DatabaseMigrationVerificationApproved
- DatabaseMigrationVerificationActivated
- DatabaseMigrationVerificationSuspended
- DatabaseMigrationVerificationCorrected
- DatabaseMigrationVerificationRetired

Reference queries:

- GetDatabaseMigrationVerification
- ListDatabaseMigrationVerificationRecords
- GetDatabaseMigrationVerificationTimeline
- GetDatabaseMigrationVerificationEvidence
- GetDatabaseMigrationVerificationDecisionPackage
- GetDatabaseMigrationVerificationExceptions
- GetDatabaseMigrationVerificationAuditTrail

Reference API surface:

```

POST /api/v1/databasemigrationverification/commands
GET  /api/v1/databasemigrationverification/{id}
GET  /api/v1/databasemigrationverification/{id}/timeline
GET  /api/v1/databasemigrationverification/{id}/evidence
GET  /api/v1/databasemigrationverification/{id}/decisions
POST /api/v1/databasemigrationverification/{id}/exceptions
POST /api/v1/databasemigrationverification/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
datasemigrationverification_aggregate	authoritative subject
datasemigrationverification_version	immutable submitted, approved, issued, or signed versions
datasemigrationverification_relationship	governed relationships
datasemigrationverification_decision	binding Decisions and rationale
datasemigrationverification_evidence_link	provenance and evidence relationships
datasemigrationverification_event_outbox	transactional Event publication
datasemigrationverification_idempotency	duplicate-write protection
datasemigrationverification_exception	exception, claim, dispute, or hold
datasemigrationverification_correction	correction and supersession lineage
datasemigrationverification_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Database Migration Verification is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Architecture Tests

Metadata	Value
Volume	17
Chapter ID	V17-07
Subject	ArchitectureTests
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Architecture Tests**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ArchitectureTests	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ArchitectureTestsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ArchitectureTestsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ArchitectureTestsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ArchitectureTestsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ArchitectureTests {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateArchitectureTests
- ValidateArchitectureTests
- ApproveArchitectureTests
- ActivateArchitectureTests
- SuspendArchitectureTests
- CorrectArchitectureTests
- RetireArchitectureTests

Reference Events:

- ArchitectureTestsCreated
- ArchitectureTestsValidated
- ArchitectureTestsApproved
- ArchitectureTestsActivated
- ArchitectureTestsSuspended
- ArchitectureTestsCorrected
- ArchitectureTestsRetired

Reference queries:

- GetArchitectureTests
- ListArchitectureTestsRecords
- GetArchitectureTestsTimeline
- GetArchitectureTestsEvidence
- GetArchitectureTestsDecisionPackage
- GetArchitectureTestsExceptions
- GetArchitectureTestsAuditTrail

Reference API surface:

```

POST /api/v1/architecturetests/commands
GET  /api/v1/architecturetests/{id}
GET  /api/v1/architecturetests/{id}/timeline
GET  /api/v1/architecturetests/{id}/evidence
GET  /api/v1/architecturetests/{id}/decisions
POST /api/v1/architecturetests/{id}/exceptions
POST /api/v1/architecturetests/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
architecturetests_aggregate	authoritative subject
architecturetests_version	immutable submitted, approved, issued, or signed versions
architecturetests_relationship	governed relationships
architecturetests_decision	binding Decisions and rationale
architecturetests_evidence_link	provenance and evidence relationships
architecturetests_event_outbox	transactional Event publication
architecturetests_idempotency	duplicate-write protection
architecturetests_exception	exception, claim, dispute, or hold
architecturetests_correction	correction and supersession lineage
architecturetests_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Architecture Tests is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED IMPLEMENTED_WITH_GAP	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION REFERENCE_ARCHITECTURE PLANNED_NOT_IMPLEMENTED DOCUMENTATION_ONLY_CLAIM	Only part of the target capability is evidenced. Normative target behavior; no claim that it exists today. Explicitly planned but not evidenced as executable. Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Tenant Isolation Tests

Metadata	Value
Volume	17
Chapter ID	V17-08
Subject	TenantIsolationTests
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Tenant Isolation Tests.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
TenantIsolationTests	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TenantIsolationTestsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TenantIsolationTestsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TenantIsolationTestsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TenantIsolationTestsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
TenantIsolationTests {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateTenantIsolationTests
- ValidateTenantIsolationTests
- ApproveTenantIsolationTests
- ActivateTenantIsolationTests
- SuspendTenantIsolationTests
- CorrectTenantIsolationTests
- RetireTenantIsolationTests

Reference Events:

- TenantIsolationTestsCreated
- TenantIsolationTestsValidated
- TenantIsolationTestsApproved
- TenantIsolationTestsActivated
- TenantIsolationTestsSuspended
- TenantIsolationTestsCorrected
- TenantIsolationTestsRetired

Reference queries:

- GetTenantIsolationTests
- ListTenantIsolationTestsRecords
- GetTenantIsolationTestsTimeline
- GetTenantIsolationTestsEvidence
- GetTenantIsolationTestsDecisionPackage
- GetTenantIsolationTestsExceptions
- GetTenantIsolationTestsAuditTrail

Reference API surface:

```

POST /api/v1/tenantisolationtests/commands
GET  /api/v1/tenantisolationtests/{id}
GET  /api/v1/tenantisolationtests/{id}/timeline
GET  /api/v1/tenantisolationtests/{id}/evidence
GET  /api/v1/tenantisolationtests/{id}/decisions
POST /api/v1/tenantisolationtests/{id}/exceptions
POST /api/v1/tenantisolationtests/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
tenantisolationtests_aggregate	authoritative subject
tenantisolationtests_version	immutable submitted, approved, issued, or signed versions
tenantisolationtests_relationship	governed relationships
tenantisolationtests_decision	binding Decisions and rationale
tenantisolationtests_evidence_link	provenance and evidence relationships
tenantisolationtests_event_outbox	transactional Event publication
tenantisolationtests_idempotency	duplicate-write protection
tenantisolationtests_exception	exception, claim, dispute, or hold
tenantisolationtests_correction	correction and supersession lineage
tenantisolationtests_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
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- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
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Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Tenant Isolation Tests is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED IMPLEMENTED_WITH_GAP	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION REFERENCE_ARCHITECTURE PLANNED_NOT_IMPLEMENTED DOCUMENTATION_ONLY_CLAIM	Only part of the target capability is evidenced. Normative target behavior; no claim that it exists today. Explicitly planned but not evidenced as executable. Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Security Adversarial Tests

Metadata	Value
Volume	17
Chapter ID	V17-09
Subject	SecurityAdversarialTests
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Security Adversarial Tests.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
SecurityAdversarialTests	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SecurityAdversarialTestsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SecurityAdversarialTestsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SecurityAdversarialTestsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SecurityAdversarialTestsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
SecurityAdversarialTests {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateSecurityAdversarialTests
- ValidateSecurityAdversarialTests
- ApproveSecurityAdversarialTests
- ActivateSecurityAdversarialTests
- SuspendSecurityAdversarialTests
- CorrectSecurityAdversarialTests
- RetireSecurityAdversarialTests

Reference Events:

- SecurityAdversarialTestsCreated
- SecurityAdversarialTestsValidated
- SecurityAdversarialTestsApproved
- SecurityAdversarialTestsActivated
- SecurityAdversarialTestsSuspended
- SecurityAdversarialTestsCorrected
- SecurityAdversarialTestsRetired

Reference queries:

- GetSecurityAdversarialTests
- ListSecurityAdversarialTestsRecords
- GetSecurityAdversarialTestsTimeline
- GetSecurityAdversarialTestsEvidence
- GetSecurityAdversarialTestsDecisionPackage
- GetSecurityAdversarialTestsExceptions
- GetSecurityAdversarialTestsAuditTrail

Reference API surface:

```

POST /api/v1/securityadversarialtests/commands
GET  /api/v1/securityadversarialtests/{id}
GET  /api/v1/securityadversarialtests/{id}/timeline
GET  /api/v1/securityadversarialtests/{id}/evidence
GET  /api/v1/securityadversarialtests/{id}/decisions
POST /api/v1/securityadversarialtests/{id}/exceptions
POST /api/v1/securityadversarialtests/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
securityadversarialtests_aggregate	authoritative subject
securityadversarialtests_version	immutable submitted, approved, issued, or signed versions
securityadversarialtests_relationship	governed relationships
securityadversarialtests_decision	binding Decisions and rationale
securityadversarialtests_evidence_link	provenance and evidence relationships
securityadversarialtests_event_outbox	transactional Event publication
securityadversarialtests_idempotency	duplicate-write protection
securityadversarialtests_exception	exception, claim, dispute, or hold
securityadversarialtests_correction	correction and supersession lineage
securityadversarialtests_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Security Adversarial Tests is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Event Replay Tests

Metadata	Value
Volume	17
Chapter ID	V17-10
Subject	EventReplayTests
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Event Replay Tests.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
EventReplayTests	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EventReplayTestsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EventReplayTestsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EventReplayTestsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EventReplayTestsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
EventReplayTests {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateEventReplayTests
- ValidateEventReplayTests
- ApproveEventReplayTests
- ActivateEventReplayTests
- SuspendEventReplayTests
- CorrectEventReplayTests
- RetireEventReplayTests

Reference Events:

- EventReplayTestsCreated
- EventReplayTestsValidated
- EventReplayTestsApproved
- EventReplayTestsActivated
- EventReplayTestsSuspended
- EventReplayTestsCorrected
- EventReplayTestsRetired

Reference queries:

- GetEventReplayTests
- ListEventReplayTestsRecords
- GetEventReplayTestsTimeline
- GetEventReplayTestsEvidence
- GetEventReplayTestsDecisionPackage
- GetEventReplayTestsExceptions
- GetEventReplayTestsAuditTrail

Reference API surface:

```

POST /api/v1/eventreplaytests/commands
GET  /api/v1/eventreplaytests/{id}
GET  /api/v1/eventreplaytests/{id}/timeline
GET  /api/v1/eventreplaytests/{id}/evidence
GET  /api/v1/eventreplaytests/{id}/decisions
POST /api/v1/eventreplaytests/{id}/exceptions
POST /api/v1/eventreplaytests/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
eventreplaytests_aggregate	authoritative subject
eventreplaytests_version	immutable submitted, approved, issued, or signed versions
eventreplaytests_relationship	governed relationships
eventreplaytests_decision	binding Decisions and rationale
eventreplaytests_evidence_link	provenance and evidence relationships
eventreplaytests_event_outbox	transactional Event publication
eventreplaytests_idempotency	duplicate-write protection
eventreplaytests_exception	exception, claim, dispute, or hold
eventreplaytests_correction	correction and supersession lineage
eventreplaytests_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Event Replay Tests is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Workflow Restart and Compensation Tests

Metadata	Value
Volume	17
Chapter ID	V17-11
Subject	WorkflowRestartandCompensationTests
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Workflow Restart and Compensation Tests**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
WorkflowRestartandCompensationTests	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WorkflowRestartandCompensationTestsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WorkflowRestartandCompensationTestsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WorkflowRestartandCompensationTestsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
WorkflowRestartandCompensationTestsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
WorkflowRestartandCompensationTests {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateWorkflowRestartandCompensationTests
- ValidateWorkflowRestartandCompensationTests
- ApproveWorkflowRestartandCompensationTests
- ActivateWorkflowRestartandCompensationTests
- SuspendWorkflowRestartandCompensationTests
- CorrectWorkflowRestartandCompensationTests
- RetireWorkflowRestartandCompensationTests

Reference Events:

- WorkflowRestartandCompensationTestsCreated
- WorkflowRestartandCompensationTestsValidated
- WorkflowRestartandCompensationTestsApproved
- WorkflowRestartandCompensationTestsActivated
- WorkflowRestartandCompensationTestsSuspended
- WorkflowRestartandCompensationTestsCorrected
- WorkflowRestartandCompensationTestsRetired

Reference queries:

- GetWorkflowRestartandCompensationTests
- ListWorkflowRestartandCompensationTestsRecords
- GetWorkflowRestartandCompensationTestsTimeline
- GetWorkflowRestartandCompensationTestsEvidence
- GetWorkflowRestartandCompensationTestsDecisionPackage
- GetWorkflowRestartandCompensationTestsExceptions
- GetWorkflowRestartandCompensationTestsAuditTrail

Reference API surface:

```

POST /api/v1/workflowrestartandcompensationtests/commands
GET  /api/v1/workflowrestartandcompensationtests/{id}
GET  /api/v1/workflowrestartandcompensationtests/{id}/timeline
GET  /api/v1/workflowrestartandcompensationtests/{id}/evidence
GET  /api/v1/workflowrestartandcompensationtests/{id}/decisions
POST /api/v1/workflowrestartandcompensationtests/{id}/exceptions
POST /api/v1/workflowrestartandcompensationtests/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C[Policy and evidence gates pass?]
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
workflowrestartandcompensationtests_aggregate	authoritative subject
workflowrestartandcompensationtests_version	immutable submitted, approved, issued, or signed versions
workflowrestartandcompensationtests_relationship	governed relationships
workflowrestartandcompensationtests_decision	binding Decisions and rationale
workflowrestartandcompensationtests_evidence_link	provenance and evidence relationships
workflowrestartandcompensationtests_event_outbox	transactional Event publication
workflowrestartandcompensationtests_idempotency	duplicate-write protection
workflowrestartandcompensationtests_exception	exception, claim, dispute, or hold
workflowrestartandcompensationtests_correction	correction and supersession lineage
workflowrestartandcompensationtests_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions

- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Workflow Restart and Compensation Tests is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

AI Evaluation

Metadata	Value
Volume	17
Chapter ID	V17-12
Subject	AIEvaluation
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for AI Evaluation.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
AIEvaluation	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AIEvaluationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AIEvaluationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AIEvaluationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AIEvaluationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```

AIEvaluation {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}

```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```



```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateAIEvaluation
- ValidateAIEvaluation
- ApproveAIEvaluation
- ActivateAIEvaluation
- SuspendAIEvaluation
- CorrectAIEvaluation
- RetireAIEvaluation

Reference Events:

- AIEvaluationCreated
- AIEvaluationValidated
- AIEvaluationApproved
- AIEvaluationActivated
- AIEvaluationSuspended
- AIEvaluationCorrected
- AIEvaluationRetired

Reference queries:

- GetAIEvaluation
- ListAIEvaluationRecords
- GetAIEvaluationTimeline
- GetAIEvaluationEvidence
- GetAIEvaluationDecisionPackage
- GetAIEvaluationExceptions
- GetAIEvaluationAuditTrail

Reference API surface:

```

POST /api/v1/aievaluation/commands
GET  /api/v1/aievaluation/{id}
GET  /api/v1/aievaluation/{id}/timeline
GET  /api/v1/aievaluation/{id}/evidence
GET  /api/v1/aievaluation/{id}/decisions
POST /api/v1/aievaluation/{id}/exceptions
POST /api/v1/aievaluation/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
aievaluation_aggregate	authoritative subject
aievaluation_version	immutable submitted, approved, issued, or signed versions
aievaluation_relationship	governed relationships
aievaluation_decision	binding Decisions and rationale
aievaluation_evidence_link	provenance and evidence relationships
aievaluation_event_outbox	transactional Event publication
aievaluation_idempotency	duplicate-write protection
aievaluation_exception	exception, claim, dispute, or hold
aievaluation_correction	correction and supersession lineage
aievaluation_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- AI Evaluation is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED IMPLEMENTED_WITH_GAP	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION REFERENCE_ARCHITECTURE PLANNED_NOT_IMPLEMENTED DOCUMENTATION_ONLY_CLAIM	Only part of the target capability is evidenced. Normative target behavior; no claim that it exists today. Explicitly planned but not evidenced as executable. Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Portal E2E Journeys

Metadata	Value
Volume	17
Chapter ID	V17-13
Subject	PortalE2EJourneys
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Portal E2E Journeys**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
PortalE2EJourneys	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PortalE2EJourneysVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PortalE2EJourneysDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PortalE2EJourneysEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PortalE2EJourneysException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
PortalE2EJourneys {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreatePortalE2EJourneys
- ValidatePortalE2EJourneys
- ApprovePortalE2EJourneys
- ActivatePortalE2EJourneys
- SuspendPortalE2EJourneys
- CorrectPortalE2EJourneys
- RetirePortalE2EJourneys

Reference Events:

- PortalE2EJourneysCreated
- PortalE2EJourneysValidated
- PortalE2EJourneysApproved
- PortalE2EJourneysActivated
- PortalE2EJourneysSuspended
- PortalE2EJourneysCorrected
- PortalE2EJourneysRetired

Reference queries:

- GetPortalE2EJourneys
- ListPortalE2EJourneysRecords
- GetPortalE2EJourneysTimeline
- GetPortalE2EJourneysEvidence
- GetPortalE2EJourneysDecisionPackage
- GetPortalE2EJourneysExceptions
- GetPortalE2EJourneysAuditTrail

Reference API surface:

```

POST /api/v1/portale2ejourneys/commands
GET  /api/v1/portale2ejourneys/{id}
GET  /api/v1/portale2ejourneys/{id}/timeline
GET  /api/v1/portale2ejourneys/{id}/evidence
GET  /api/v1/portale2ejourneys/{id}/decisions
POST /api/v1/portale2ejourneys/{id}/exceptions
POST /api/v1/portale2ejourneys/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```



```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
portale2ejourneys_aggregate	authoritative subject
portale2ejourneys_version	immutable submitted, approved, issued, or signed versions
portale2ejourneys_relationship	governed relationships
portale2ejourneys_decision	binding Decisions and rationale
portale2ejourneys_evidence_link	provenance and evidence relationships
portale2ejourneys_event_outbox	transactional Event publication
portale2ejourneys_idempotency	duplicate-write protection
portale2ejourneys_exception	exception, claim, dispute, or hold
portale2ejourneys_correction	correction and supersession lineage
portale2ejourneys_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Portal E2E Journeys is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Accessibility Testing

Metadata	Value
Volume	17
Chapter ID	V17-14
Subject	AccessibilityTesting
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Accessibility Testing**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
AccessibilityTesting	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AccessibilityTestingVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AccessibilityTestingDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AccessibilityTestingEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AccessibilityTestingException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
AccessibilityTesting {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateAccessibilityTesting
- ValidateAccessibilityTesting
- ApproveAccessibilityTesting
- ActivateAccessibilityTesting
- SuspendAccessibilityTesting
- CorrectAccessibilityTesting
- RetireAccessibilityTesting

Reference Events:

- AccessibilityTestingCreated
- AccessibilityTestingValidated
- AccessibilityTestingApproved
- AccessibilityTestingActivated
- AccessibilityTestingSuspended
- AccessibilityTestingCorrected
- AccessibilityTestingRetired

Reference queries:

- GetAccessibilityTesting
- ListAccessibilityTestingRecords
- GetAccessibilityTestingTimeline
- GetAccessibilityTestingEvidence
- GetAccessibilityTestingDecisionPackage
- GetAccessibilityTestingExceptions
- GetAccessibilityTestingAuditTrail

Reference API surface:

```

POST /api/v1/accessibilitytesting/commands
GET  /api/v1/accessibilitytesting/{id}
GET  /api/v1/accessibilitytesting/{id}/timeline
GET  /api/v1/accessibilitytesting/{id}/evidence
GET  /api/v1/accessibilitytesting/{id}/decisions
POST /api/v1/accessibilitytesting/{id}/exceptions
POST /api/v1/accessibilitytesting/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
accessibilitytesting_aggregate	authoritative subject
accessibilitytesting_version	immutable submitted, approved, issued, or signed versions
accessibilitytesting_relationship	governed relationships
accessibilitytesting_decision	binding Decisions and rationale
accessibilitytesting_evidence_link	provenance and evidence relationships
accessibilitytesting_event_outbox	transactional Event publication
accessibilitytesting_idempotency	duplicate-write protection
accessibilitytesting_exception	exception, claim, dispute, or hold
accessibilitytesting_correction	correction and supersession lineage
accessibilitytesting_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Accessibility Testing is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Performance and Capacity Testing

Metadata	Value
Volume	17
Chapter ID	V17-15
Subject	PerformanceandCapacityTesting
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Performance and Capacity Testing**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
PerformanceandCapacityTesting	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PerformanceandCapacityTestingVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PerformanceandCapacityTestingDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PerformanceandCapacityTestingEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PerformanceandCapacityTestingException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
PerformanceandCapacityTesting {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreatePerformanceandCapacityTesting
- ValidatePerformanceandCapacityTesting
- ApprovePerformanceandCapacityTesting
- ActivatePerformanceandCapacityTesting
- SuspendPerformanceandCapacityTesting
- CorrectPerformanceandCapacityTesting
- RetirePerformanceandCapacityTesting

Reference Events:

- PerformanceandCapacityTestingCreated
- PerformanceandCapacityTestingValidated
- PerformanceandCapacityTestingApproved
- PerformanceandCapacityTestingActivated
- PerformanceandCapacityTestingSuspended
- PerformanceandCapacityTestingCorrected
- PerformanceandCapacityTestingRetired

Reference queries:

- GetPerformanceandCapacityTesting
- ListPerformanceandCapacityTestingRecords
- GetPerformanceandCapacityTestingTimeline
- GetPerformanceandCapacityTestingEvidence
- GetPerformanceandCapacityTestingDecisionPackage
- GetPerformanceandCapacityTestingExceptions
- GetPerformanceandCapacityTestingAuditTrail

Reference API surface:

```

POST /api/v1/performanceandcapacitytesting/commands
GET  /api/v1/performanceandcapacitytesting/{id}
GET  /api/v1/performanceandcapacitytesting/{id}/timeline
GET  /api/v1/performanceandcapacitytesting/{id}/evidence
GET  /api/v1/performanceandcapacitytesting/{id}/decisions
POST /api/v1/performanceandcapacitytesting/{id}/exceptions
POST /api/v1/performanceandcapacitytesting/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
performanceandcapacitytesting_aggregate	authoritative subject
performanceandcapacitytesting_version	immutable submitted, approved, issued, or signed versions
performanceandcapacitytesting_relationship	governed relationships
performanceandcapacitytesting_decision	binding Decisions and rationale
performanceandcapacitytesting_evidence_link	provenance and evidence relationships
performanceandcapacitytesting_event_outbox	transactional Event publication
performanceandcapacitytesting_idempotency	duplicate-write protection
performanceandcapacitytesting_exception	exception, claim, dispute, or hold
performanceandcapacitytesting_correction	correction and supersession lineage
performanceandcapacitytesting_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Performance and Capacity Testing is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Resilience and Fault Injection

Metadata	Value
Volume	17
Chapter ID	V17-16
Subject	ResilienceandFaultInjection
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Resilience and Fault Injection**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ResilienceandFaultInjection	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ResilienceandFaultInjectionVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ResilienceandFaultInjectionDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ResilienceandFaultInjectionEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ResilienceandFaultInjectionException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ResilienceandFaultInjection {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateResilienceandFaultInjection
- ValidateResilienceandFaultInjection
- ApproveResilienceandFaultInjection
- ActivateResilienceandFaultInjection
- SuspendResilienceandFaultInjection
- CorrectResilienceandFaultInjection
- RetireResilienceandFaultInjection

Reference Events:

- ResilienceandFaultInjectionCreated
- ResilienceandFaultInjectionValidated
- ResilienceandFaultInjectionApproved
- ResilienceandFaultInjectionActivated
- ResilienceandFaultInjectionSuspended
- ResilienceandFaultInjectionCorrected
- ResilienceandFaultInjectionRetired

Reference queries:

- GetResilienceandFaultInjection
- ListResilienceandFaultInjectionRecords
- GetResilienceandFaultInjectionTimeline
- GetResilienceandFaultInjectionEvidence
- GetResilienceandFaultInjectionDecisionPackage
- GetResilienceandFaultInjectionExceptions
- GetResilienceandFaultInjectionAuditTrail

Reference API surface:

```

POST /api/v1/resilienceandfaultinjection/commands
GET  /api/v1/resilienceandfaultinjection/{id}
GET  /api/v1/resilienceandfaultinjection/{id}/timeline
GET  /api/v1/resilienceandfaultinjection/{id}/evidence
GET  /api/v1/resilienceandfaultinjection/{id}/decisions
POST /api/v1/resilienceandfaultinjection/{id}/exceptions
POST /api/v1/resilienceandfaultinjection/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
resilienceandfaultinjection_aggregate	authoritative subject
resilienceandfaultinjection_version	immutable submitted, approved, issued, or signed versions
resilienceandfaultinjection_relationship	governed relationships
resilienceandfaultinjection_decision	binding Decisions and rationale
resilienceandfaultinjection_evidence_link	provenance and evidence relationships
resilienceandfaultinjection_event_outbox	transactional Event publication
resilienceandfaultinjection_idempotency	duplicate-write protection
resilienceandfaultinjection_exception	exception, claim, dispute, or hold
resilienceandfaultinjection_correction	correction and supersession lineage
resilienceandfaultinjection_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Resilience and Fault Injection is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Disaster Recovery Exercises

Metadata	Value
Volume	17
Chapter ID	V17-17
Subject	DisasterRecoveryExercises
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Disaster Recovery Exercises**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DisasterRecoveryExercises	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DisasterRecoveryExercisesVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DisasterRecoveryExercisesDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DisasterRecoveryExercisesEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DisasterRecoveryExercisesException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DisasterRecoveryExercises {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDisasterRecoveryExercises
- ValidateDisasterRecoveryExercises
- ApproveDisasterRecoveryExercises
- ActivateDisasterRecoveryExercises
- SuspendDisasterRecoveryExercises
- CorrectDisasterRecoveryExercises
- RetireDisasterRecoveryExercises

Reference Events:

- DisasterRecoveryExercisesCreated
- DisasterRecoveryExercisesValidated
- DisasterRecoveryExercisesApproved
- DisasterRecoveryExercisesActivated
- DisasterRecoveryExercisesSuspended
- DisasterRecoveryExercisesCorrected
- DisasterRecoveryExercisesRetired

Reference queries:

- GetDisasterRecoveryExercises
- ListDisasterRecoveryExercisesRecords
- GetDisasterRecoveryExercisesTimeline
- GetDisasterRecoveryExercisesEvidence
- GetDisasterRecoveryExercisesDecisionPackage
- GetDisasterRecoveryExercisesExceptions
- GetDisasterRecoveryExercisesAuditTrail

Reference API surface:

```

POST /api/v1/disasterrecoveryexercises/commands
GET  /api/v1/disasterrecoveryexercises/{id}
GET  /api/v1/disasterrecoveryexercises/{id}/timeline
GET  /api/v1/disasterrecoveryexercises/{id}/evidence
GET  /api/v1/disasterrecoveryexercises/{id}/decisions
POST /api/v1/disasterrecoveryexercises/{id}/exceptions
POST /api/v1/disasterrecoveryexercises/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
disasterrecoveryexercises_aggregate	authoritative subject
disasterrecoveryexercises_version	immutable submitted, approved, issued, or signed versions
disasterrecoveryexercises_relationship	governed relationships
disasterrecoveryexercises_decision	binding Decisions and rationale
disasterrecoveryexercises_evidence_link	provenance and evidence relationships
disasterrecoveryexercises_event_outbox	transactional Event publication
disasterrecoveryexercises_idempotency	duplicate-write protection
disasterrecoveryexercises_exception	exception, claim, dispute, or hold
disasterrecoveryexercises_correction	correction and supersession lineage
disasterrecoveryexercises_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Disaster Recovery Exercises is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Partner Certification

Metadata	Value
Volume	17
Chapter ID	V17-18
Subject	PartnerCertification
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Partner Certification**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
PartnerCertification	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PartnerCertificationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PartnerCertificationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PartnerCertificationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PartnerCertificationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
PartnerCertification {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreatePartnerCertification
- ValidatePartnerCertification
- ApprovePartnerCertification
- ActivatePartnerCertification
- SuspendPartnerCertification
- CorrectPartnerCertification
- RetirePartnerCertification

Reference Events:

- PartnerCertificationCreated
- PartnerCertificationValidated
- PartnerCertificationApproved
- PartnerCertificationActivated
- PartnerCertificationSuspended
- PartnerCertificationCorrected
- PartnerCertificationRetired

Reference queries:

- GetPartnerCertification
- ListPartnerCertificationRecords
- GetPartnerCertificationTimeline
- GetPartnerCertificationEvidence
- GetPartnerCertificationDecisionPackage
- GetPartnerCertificationExceptions
- GetPartnerCertificationAuditTrail

Reference API surface:

```

POST /api/v1/partnercertification/commands
GET  /api/v1/partnercertification/{id}
GET  /api/v1/partnercertification/{id}/timeline
GET  /api/v1/partnercertification/{id}/evidence
GET  /api/v1/partnercertification/{id}/decisions
POST /api/v1/partnercertification/{id}/exceptions
POST /api/v1/partnercertification/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
partnercertification_aggregate	authoritative subject
partnercertification_version	immutable submitted, approved, issued, or signed versions
partnercertification_relationship	governed relationships
partnercertification_decision	binding Decisions and rationale
partnercertification_evidence_link	provenance and evidence relationships
partnercertification_event_outbox	transactional Event publication
partnercertification_idempotency	duplicate-write protection
partnercertification_exception	exception, claim, dispute, or hold
partnercertification_correction	correction and supersession lineage
partnercertification_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Partner Certification is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Continuous Conformance

Metadata	Value
Volume	17
Chapter ID	V17-19
Subject	ContinuousConformance
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define the target architecture, implementation standard, evidence, security, reliability, operations, tests, migration and conformance requirements for Continuous Conformance.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ContinuousConformance	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ContinuousConformanceVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ContinuousConformanceDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ContinuousConformanceEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ContinuousConformanceException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ContinuousConformance {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateContinuousConformance
- ValidateContinuousConformance
- ApproveContinuousConformance
- ActivateContinuousConformance
- SuspendContinuousConformance
- CorrectContinuousConformance
- RetireContinuousConformance

Reference Events:

- ContinuousConformanceCreated
- ContinuousConformanceValidated
- ContinuousConformanceApproved
- ContinuousConformanceActivated
- ContinuousConformanceSuspended
- ContinuousConformanceCorrected
- ContinuousConformanceRetired

Reference queries:

- GetContinuousConformance
- ListContinuousConformanceRecords
- GetContinuousConformanceTimeline
- GetContinuousConformanceEvidence
- GetContinuousConformanceDecisionPackage
- GetContinuousConformanceExceptions
- GetContinuousConformanceAuditTrail

Reference API surface:

```

POST /api/v1/continuousconformance/commands
GET  /api/v1/continuousconformance/{id}
GET  /api/v1/continuousconformance/{id}/timeline
GET  /api/v1/continuousconformance/{id}/evidence
GET  /api/v1/continuousconformance/{id}/decisions
POST /api/v1/continuousconformance/{id}/exceptions
POST /api/v1/continuousconformance/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
continuousconformance_aggregate	authoritative subject
continuousconformance_version	immutable submitted, approved, issued, or signed versions
continuousconformance_relationship	governed relationships
continuousconformance_decision	binding Decisions and rationale
continuousconformance_evidence_link	provenance and evidence relationships
continuousconformance_event_outbox	transactional Event publication
continuousconformance_idempotency	duplicate-write protection
continuousconformance_exception	exception, claim, dispute, or hold
continuousconformance_correction	correction and supersession lineage
continuousconformance_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions
- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- Continuous Conformance is part of the target GTOS handbook scope.
- Exact implementation, configuration, deployment, tests and runtime evidence require repository and environment inspection.
- Plans and infrastructure manifests are classified separately from observed runtime behavior.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED IMPLEMENTED_WITH_GAP	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION REFERENCE_ARCHITECTURE PLANNED_NOT_IMPLEMENTED DOCUMENTATION_ONLY_CLAIM	Only part of the target capability is evidenced. Normative target behavior; no claim that it exists today. Explicitly planned but not evidenced as executable. Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Volume 17 Conformance and Evidence Matrix

Metadata	Value
Volume	17
Chapter ID	V17-SYNTH
Subject	Volume17ConformanceMatrix
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Quality Engineering Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **trace every requirement in Volume 17 to owner, repository source, configuration, deployment, runtime evidence, tests, runbooks, exceptions and review status**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome
≠ Observation
≠ Claim
≠ derived Perspective
≠ Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Quality Engineering Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
Volume17ConformanceMatrix	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Volume17ConformanceMatrixVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Volume17ConformanceMatrixDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Volume17ConformanceMatrixEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Volume17ConformanceMatrixException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
Volume17ConformanceMatrix {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```



```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateVolume17ConformanceMatrix
- ValidateVolume17ConformanceMatrix
- ApproveVolume17ConformanceMatrix
- ActivateVolume17ConformanceMatrix
- SuspendVolume17ConformanceMatrix
- CorrectVolume17ConformanceMatrix
- RetireVolume17ConformanceMatrix

Reference Events:

- Volume17ConformanceMatrixCreated
- Volume17ConformanceMatrixValidated
- Volume17ConformanceMatrixApproved
- Volume17ConformanceMatrixActivated
- Volume17ConformanceMatrixSuspended
- Volume17ConformanceMatrixCorrected
- Volume17ConformanceMatrixRetired

Reference queries:

- GetVolume17ConformanceMatrix
- ListVolume17ConformanceMatrixRecords
- GetVolume17ConformanceMatrixTimeline
- GetVolume17ConformanceMatrixEvidence
- GetVolume17ConformanceMatrixDecisionPackage
- GetVolume17ConformanceMatrixExceptions
- GetVolume17ConformanceMatrixAuditTrail

Reference API surface:

```

POST /api/v1/volumel7conformancematrix/commands
GET  /api/v1/volumel7conformancematrix/{id}
GET  /api/v1/volumel7conformancematrix/{id}/timeline
GET  /api/v1/volumel7conformancematrix/{id}/evidence
GET  /api/v1/volumel7conformancematrix/{id}/decisions
POST /api/v1/volumel7conformancematrix/{id}/exceptions
POST /api/v1/volumel7conformancematrix/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
volume17conformancematrix_aggregate	authoritative subject
volume17conformancematrix_version	immutable submitted, approved, issued, or signed versions
volume17conformancematrix_relationship	governed relationships
volume17conformancematrix_decision	binding Decisions and rationale
volume17conformancematrix_evidence_link	provenance and evidence relationships
volume17conformancematrix_event_outbox	transactional Event publication
volume17conformancematrix_idempotency	duplicate-write protection
volume17conformancematrix_exception	exception, claim, dispute, or hold
volume17conformancematrix_correction	correction and supersession lineage
volume17conformancematrix_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions

- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.